

Project MediLong

ML Fusion & Clinical Safety Squad — Detailed Report

Squad Report · 27-Day Sprint · Sept 2 – Sept 28

Squad	ML Fusion & Clinical Safety
Members	Manoj, Nambi, Deepak, (4th member TBD)
Sprint Window	Sept 2 – Sept 28, 2026 (27-day sprint)

Mission

Build the predictive core: the XGBoost risk classifier, the deterministic clinical safety guardrails, and the SHAP explainability layer. This squad decides what the system actually diagnoses and defends those decisions.

Contract Slice Owned

decision_core (risk_probability, risk_category, safety_override, shap_contributions).

Sprint-by-Sprint Tasks

Sprint 1 (Sept 2–8)

- Load and clean the Kaggle Indian PCOS dataset (or equivalent Rotterdam-compliant labeled data).
- Baseline XGBoost training run — establish an accuracy/F1 floor to improve against.
- Define the deterministic guardrail rule table from FOGSI/Rotterdam criteria (e.g. TSH > 4.5 uIU/mL → hypothyroidism differential alert, overriding a PCOS-positive prediction).

Sprint 2 (Sept 9–15)

- Finalize XGBoost model with probability calibration (e.g. isotonic regression) so risk_probability is meaningfully continuous, not just a raw score.
- Integrate SHAP TreeExplainer; define the shap_contributions payload shape for Frontend's waterfall chart.
- Implement guardrail layer as a pre-model override check — it must run and can veto before XGBoost's output is trusted.

Sprint 3 (Sept 16–22)

- Integrate with Backend's /predict to consume real Pillar 1/2/3 data (Pillar 2 in stub mode) and return decision_core live.
- Validate that stub-mode CV data doesn't silently skew the model — consider down-weighting or flagging Pillar 2 features when mode: "stub".

Sprint 4 (Sept 23–28)

- Edge-case validation: borderline biomarker values, conflicting guardrail vs. model signals, missing-field inputs.
- Confirm guardrail overrides are unit-tested independently of the XGBoost path.

Dependencies

- Blocked by: Data Pipelines (needs real biomarker output by Sprint 2 to validate against), Backend (integration point).
- Blocks: Frontend (SHAP payload shape), Backend (PDF summary content).

Definition of Done

- Guardrail layer demonstrably overrides model output on at least one hard clinical rule (e.g. TSH override) in testing.
- SHAP output renders correctly in Frontend's waterfall chart.
- Risk categorization stable and reproducible across repeated runs on the same input.